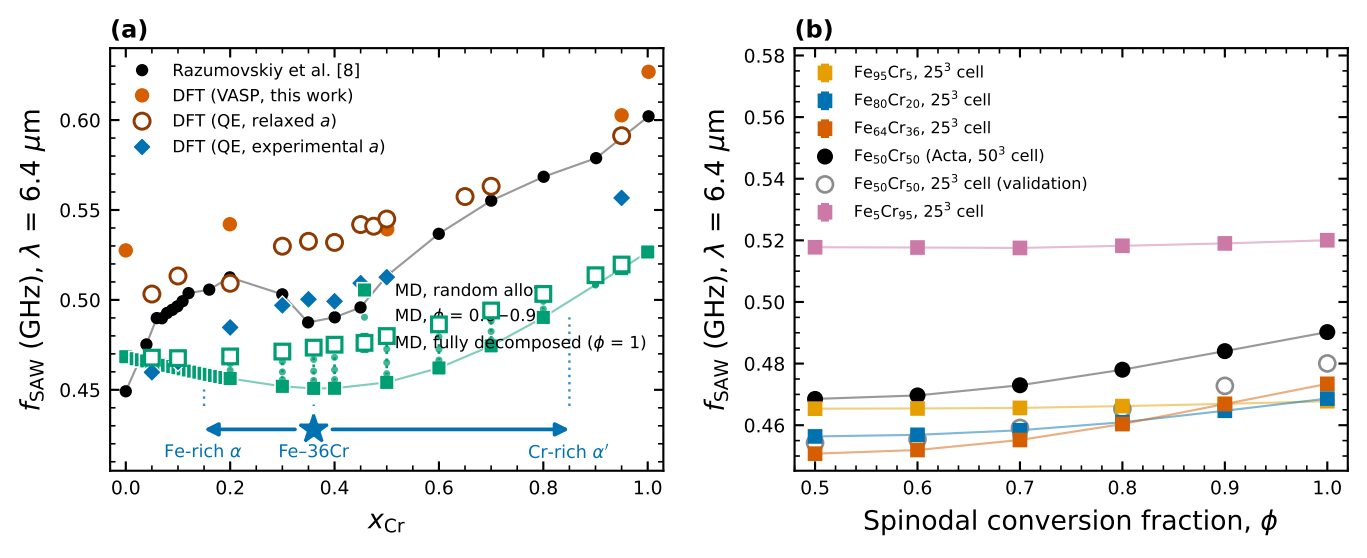
(a)
Razumovskiy et al. [8]
DFT (VASP, this work)
DFT (QE, relaxed a)
DFT (QE, experimental a)
MD, random alloy
MD, ϕ = 0.1–0.9
MD, fully decomposed (ϕ = 1)
Fe-rich α
Fe–36Cr
Cr-rich α′
f_{SAW} (GHz), λ = 6.4 μm
x_{Cr}
0.0
0.2
0.4
0.6
0.8
1.0
0.45
0.50
0.55
0.60
(b)
$Fe_{95}Cr_5$, 25^3 cell
$Fe_{80}Cr_{20}$, 25^3 cell
$Fe_{64}Cr_{36}$, 25^3 cell
$Fe_{50}Cr_{50}$ (Acta, 50^3 cell)
$Fe_{50}Cr_{50}$, 25^3 cell (validation)
Fe_5Cr_{95}, 25^3 cell
f_{SAW} (GHz), λ = 6.4 μm
Spinodal conversion fraction, ϕ
0.5
0.6
0.7
0.8
0.9
1.0
0.46
0.48
0.50
0.52
0.54
0.56
0.58